

Roy J. Harnish

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Professional Experience: Software Development, Medical Image Analysis, Scientific Computing, Machine and Deep Learning

Vail Resorts, Winter 23/24 – 24/25

- Ski Instructor at Northstar Ski and Ride School.

Curia LLC, Aug. 2017 – June 2023

- Software/Machine-Learning engineer for a biometric analysis firm seeking to characterize human performance in eSports. Worked with Founder/CEO to define objectives and architect solutions. Created software to stream data from a collection of biometric sensing devices (EEG, PPG, Eyetracking) to LabStreamingLayer and tools to fuse data from asynchronous sources (game stream capture and LSL). Developed analysis and reporting software in Python. Constructed image processing pipelines using OpenCV GPU and C++ to label video game footage for ML/DL. Developed or selected ML/DL models to classify events in images and time-series data using TensorFlow Keras. Created GUI applications to provide client-side telemetry and to interface with recording, reporting, and analysis software in WPF and WinForms using C#.

ProxSim, July 2017 – Sept. 2017

- Software consultant to a design firm that produced a surgical simulation software for a medical implant company. Helped to develop and choose approaches for DICOM/PACS integration into their product, and implemented a rotating planes viewer for 3D image volumes in javascript.

UCSF, Department of Radiology, Franc Lab, July 2015 – June 2017

- NeuroPET Deep Learning -- Built classifier of Neuro PET images for differentiation between Normal, Mild Cognitive Impairment, and Alzheimer's. Constructed pipeline to pull from PACS, anonymize, and process, DICOM images from FDG-PET studies of the brain into sets of homologous slices. Obtained and processed Analyze format images from ADNI web portal. Trained convolutional neural network (CNN) model using Keras library. Used ADNI for train/validation and UCSF data as test set.
- Mammography Deep Learning -- Developed a pipeline to pull from PACS and anonymize mammography images, and to extract BIRADS scores from radiology reports via NLP techniques. Obtained pathology reports and integrated data from pathology and radiology reports with images.
- PETCTomics -- Provided programming support for an application of Radiomics to integrated whole body PETCT and breast MRI. Built NoSQL (MongoDB) database to keep track of image processing stages and to store derived data. Created custom modules to interface with ITK Python bindings in LPS frame of reference.
- Chemo brain -- Built pipeline to automatically localize the brain in whole body PETCT images and perform Atlas-based segmentation. Extracted metrics using ROIs obtained via segmentation to generate tabular data for statistical analysis.
- Augmented Reality -- Worked on various prototypes for augmented reality app to view 3D medical image data. Experimented with Google Glass and Android tablets as hardware platforms. Used Marker Based Tracking library to perform virtual camera pose estimation.

UCSF, Department of Radiology, Breast Imaging Research Program, July 2013 – Nov. 2016

- Breast MRI Registration -- Developed image registration software modules for registration of diffusion and DCE MRI images of the breast and integrated modules into large in house breast MRI processing software.
- Whole Breast Segmentation -- Developed software for automated segmentation of whole breast tissue from MRI images.
- BPE Visualization -- Built tool for visualization of background parenchymal enhancement (BPE) in breast MRI.
- SER Volume recovery from PDF reports -- Created software to align MIPs of breast MRI images to PDF reports to recover annotations.

UCSF, Department of Radiology, Lang Lab, July 2009 – Mar. 2015

- MIPAV QCT -- Ported in-house Quantitative Computed Tomography (QCT) analysis software to suite of MIPAV (Java) plugins. The existing software was comprised of shell scripts, AVS networks, and C-modules, and ran on a solitary Red Hat Linux machine. The objective was to translate the software, which performs analysis of muscle and bone in QCT images of the hip and spine, into a series of plugins to the Java-based MIPAV, an image processing environment developed by the NIH. The control flow and image interaction performed by the scripts and AVS networks was replaced with MIPAV plugins, and the C-modules were modified and compiled to run on all major operating systems. This allows the software to run on any machine supported by MIPAV (which covers all major operating systems). I used the eclipse IDE for development, and subclipse for SVN version control.
- CT MAR for PET AC -- Primary researcher on a project involving the development and application of CT metal artifact reduction (MAR) methods to the improvement of PET attenuation correction in the vicinity of metallic hip implants. Wrote MATLAB code that performs MAR on PET/CT data. Used Python to glue together PET reconstruction binaries to allow for PET image reconstruction on a PC workstation.
- C-11 Methionine Protein Synthesis -- Developed a data analysis pipeline for a dynamic PET study of C-11 methionine kinetics in skeletal muscle tissue following unilateral resistance exercise by subjects. Wrote programs to; retrieve DICOM data from our network server, have radiologists mark anatomical fiducials in CT scout images, map these markers to axial sections of CT and PET images, automatically segment muscle tissue from CT axial slices, register PET images to CT images, segment femoral arteries from dynamic PET images, deconvolve PET images to allow an arterial input function to be determined from the images (where it was previously necessary to place an arterial catheter in subjects), get time activity curves from segmented muscle in the PET images, and export all required data for analysis on PKIN kinetic analysis software (PMOD Technologies, Zurich, Switzerland). Created MeVisLab networks to produce visualizations of dynamic PET data and PET/CT fusion images.
- MATLAB QCT -- Modified QCT vertebral analysis software to allow cases from several unusable data sets to be analyzed. Re-worked the software in its original combination of C and BASH, and rewrote a section in MATLAB, producing a simple user-friendly GUI, which allowed for thousands of more cases to be incorporated into our collaborator's study.

UCSF, Department of Radiology, Prevrhal Lab, Mar. 2008 – July 2009

- Medical Image Augmented Reality -- Developed multiple prototypes of a system for viewing medical images superimposed over real-time video and displayed on a tablet PC. Early prototypes integrated two Bluetooth IR cameras and IR emitting LEDs to estimate the device's spatial position and orientation (pose) relative to a CT phantom, and a USB camera to acquire video. I used .NET

wrappers of VTK for volume rendering, and a .NET library for the IR cameras, and wrote the software in C# using the Visual Studio IDE. Later prototypes offset the device's pose estimation to an augmented reality marker based tracking system.

- CT Angiography Blood Flow Velocimetry -- Worked with collaborators to design, implement, and analyze results from -- an experiment to measure blood flow velocity using raw CT projection data. Wrote software to analyze high speed video of our experiments, and worked with another physicist on scanner geometry and code for analysis of CT projection data. Our results were published in Radiology.
- Quantitative Morphometry -- Project Manager of a digital x-ray analysis study. Modified and debugged our Quantitative Morphometry (QM) software, which used a Tcl/Tk VTK user interface, Python control flow, Conquest DICOM server, and MySQL database. Delivered SQL scripts for database management, and SAS scripts to export data for statistical analysis.

UCSF, Department of Radiation Oncology, Helen Diller Family Comprehensive Cancer Center -- Faddegon Lab, Sept. 2006 – Nov. 2007

- F.A.S.T. (Master's Thesis) -- Conducted research, and wrote Master's Thesis: "The Feasibility of a Final Aperture Superposition Technique for X-ray Radiotherapy." This consisted of a series of numerical experiments in which we modeled radiation transport in a clinical linear accelerator to estimate the radiological dose deposition in a water phantom. We used the BEAMnrc Monte Carlo Simulation System and a cluster of Linux machines to perform these calculations.

Education:

M.S., Physics, 2009 -- San Francisco State University, San Francisco, CA

B.A., Comparative Literature, 2001 -- University of Michigan, Ann Arbor, MI

Publications and Proceedings:

Yiming Ding, Jae Ho Sohn, MD, MS, Michael G. Kawczynski, MS, Hari Trivedi, MD, Roy Harnish, MS, Nathaniel W. Jenkins, MS, Dmytro Lituiev, PhD, Timothy P. Copeland, MPP, Mariam S. Aboian, MD, PhD, Carina Mari Aparici, MD, Spencer C. Behr, MD, Robert R. Flavell, MD, PhD, Shih-Ying Huang, PhD, Kelly A. Zalocusky, PhD, Lorenzo Nardo, PhD, Youngho Seo, PhD, Randall A. Hawkins, MD, PhD, Miguel Hernandez Pampaloni, MD, PhD, Dexter Hadley, MD, PhD, Benjamin L. Franc, MD, MS. "A Deep Learning Model to Predict a Diagnosis of Alzheimer Disease by Using 18F-FDG PET of the Brain," Radiology (2018)

Dmytro S. Lituiev, Hari Trivedi, Maryam Panahiazar, Beau Norgeot, Youngho Seo, Benjamin Franc, Roy Harnish, Michael Kawczynski, Dexter Hadley, "Automatic Labeling of Special Diagnostic Mammography Views from Images and DICOM Headers," Journal of Digital Imaging (2018)

Shih-ying Huang, Benjamin L. Franc, Roy J. Harnish, Gengbo Liu, Debasis Mitra, Timothy P. Copeland, Vignesh A. Arasu, John Kornak, Ella F. Jones, Spencer C. Behr, Nola M. Hylton, Elissa R. Price, Laura Esserman, Youngho Seo, "Exploration of PET and MRI radiomic features for decoding breast cancer phenotypes and prognosis," NPJCancer (2018)

Arentson-Lantz EJ, Saeed IH, Frassetto LA, Masharani U, Harnish RJ, Seo Y, VanBrocklin HF, Hawkins RA, Mari-Aparici C, Pampaloni MH, Slater J, Paddon-Jones D, Lang TF. "11C-L-methyl methionine dynamic PET/CT of skeletal muscle: response to protein supplementation compared to L-[ring 13C6] phenylalanine infusion with serial muscle biopsy," Ann Nucl Med (2017)

Julio Carballido-Gamio, Serena Bonaretti, Isra Saeed, Roy Harnish, Robert Recker, Andrew J. Burghardt, Joyce H. Keyak, Tamara Harris, Sundeep Khosla, Thomas F. Lang. "Automatic multi-parametric

quantification of the proximal femur with quantitative computed tomography," Quantitative Imaging in Medicine and Surgery (2015)

Heilmeier U, Carpenter DR, Patsch JM, Harnish R, Joseph GB, Burghardt AJ, Baum T, Schwartz AV, Lang TF, Link TM. "Volumetric femoral BMD, bone geometry, and serum sclerostin levels differ between type 2 diabetic postmenopausal women with and without fragility fractures," Osteoporos Int. (2015)

Harnish R, Prevrhal S, Alavi A, Zaidi H, Lang TF. "The effect of metal artefact reduction on CT-based attenuation correction for PET imaging in the vicinity of metallic hip implants: a phantom study," Ann Nucl Med (2014)

Thomas F. Lang, Isra H. Saeed, Timothy Streeper, Julio Carballido-Gamio, Roy J. Harnish, Lynda A. Frassetto, Stuart M.C. Lee, Jean D. Sibonga, Joyce H. Keyak, Barry A. Spiering, Carlos M. Grodzinsky, Jacob J. Bloomberg, Peter R. Cavanagh. "Spatial Heterogeneity In The Response Of The Proximal Femur To Two Lower-Body Resistance Exercise Regimens," J Bone Miner Res (2014)

Julio Carballido-Gamio, Ph.D., Roy Harnish, M.S., Isra Saeed, M.D., Timothy Streeper, M.S., Sigurdur Sigurdsson, Ph.D., Shreyasee Amin, M.D., Elizabeth J. Atkinson, M.S., Terry M. Therneau, Ph.D., Kristin Siggeirsdottir, M.S., Xiaoguang Cheng, M.D., L. Joseph Melton III, M.D., Joyce Keyak, Ph.D., Vilmundur Gudnason, M.D., Sundeep Khosla, M.D., Tamara B. Harris, M.D., and Thomas F. Lang, Ph.D. "Structural patterns of the proximal femur in relation to age and hip fracture risk in women," Bone (2013)

Julio Carballido-Gamio, Ph.D., Roy Harnish, M.S., Isra Saeed, M.D., Timothy Streeper, M.S., Sigurdur Sigurdsson, Ph.D., Shreyasee Amin, M.D., Elizabeth J. Atkinson, M.S., Terry M. Therneau, Ph.D., Kristin Siggeirsdottir, M.S., Xiaoguang Cheng, M.D., L. Joseph Melton III, M.D., Joyce Keyak, Ph.D., Vilmundur Gudnason, M.D., Sundeep Khosla, M.D., Tamara B. Harris, M.D., Thomas F. Lang, Ph.D. "Proximal femoral density distribution and structure in relation to age and hip fracture risk in women," J Bone Miner Res. (2013)

R. Harnish, T. Streeper, I. Saeed, C. Schreck, S. Dannoon, J. Slater, J. Blecha, H. VanBrocklin, M. Hernandez-Pampaloni, R. Hawkins, Y. Seo, G. Sayre, T. Lang, "Quantification of changes in skeletal muscle amino acid kinetics in adult humans in response to exercise via positron-emission tomography with L-[methyl-11C]methionine," Journal of Molecular Imaging and Dynamics (2012)

Sven Prevrhal, Carlos H. Forsythe, Roy J. Harnish, Maythem Saeed, Benjamin M. Yeh, "CT Angiographic Measurement of Vascular Blood Flow Velocity by Using Projection Data," Radiology 261, 923-929 (2011)

Roy Harnish MS, Lorenzo Nardo MD, Timothy Copeland MPP, Youngho Seo PhD, Carina Mari Aparici MD, and Benjamin Franc MD "Regional ¹⁸F-fluorodeoxyglucose (FDG) uptake in the brain on whole body PET-CT of breast cancer patients receiving chemotherapy" (SNMMI 2016 Poster)